Huawei Access Node BAS-ACC-73 Optical Transport Failure SOP (ALARM_1249)

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1249

Description

This Standard Operating Procedure outlines the steps to diagnose, resolve, and verify the resolution of an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node BAS-ACC-73. This failure domain is specific to the Access role in the Basavanagudi POP, where the Optical Transport Network (OTN) is responsible for transmitting data between the node and the Optical Line Terminal (OLT) or Optical Distribution Frame (ODF). The OTN protocol stack is based on the Synchronous Optical Networking (SONET) and Synchronous Digital Hierarchy (SDH) standards, which utilize a 155.52 Mbps synchronous digital signal to transport data. The SFP/XFP transceivers, which are used for optical transmission, can degrade over time due to physical stress, temperature changes, or optical interference. Additionally, fiber cuts or Rx/Tx power loss can also cause this failure domain. The ALARM_1249 alarm code indicates a warning-level issue, which requires prompt attention to prevent network downtime and data loss.

Pre-requisite Checks

- check: Verify the power status of the node and ensure it is in the correct state to perform the diagnostic and resolution steps.

- check: Check the fiber connections to the SFP/XFP transceivers for any signs of damage or degradation.

- check: Verify the software version of the node and ensure it is up-to-date with the latest patches.

- check: Check the redundant link status to ensure that it is functioning correctly.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP transceiver	High
Fiber cut or Rx/Tx power loss	Medium
Software or firmware issue	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver, Expected Output: Transceiver information, including temperature, voltage, and optical power levels., Description: This command displays the transceiver information, which can help identify any issues with the SFP/XFP transceivers.
```

```
Command: display optical-link, Expected Output: Optical link information, including receive and transmit power levels, and fiber status., Description: This command displays the optical link information, which can help identify any issues with the fiber or Rx/Tx power loss.

Command: display system, Expected Output: System information, including software version and redundant link status., Description: This command displays the system information, which can help identify any software or firmware issues.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceiver, Command: replace transceiver , Description: Replace the degraded SFP/XFP transceiver with a new one to resolve the issue.

Step 2: Step: Repair or replace the fiber, Command: repair fiber or replace fiber , Description: Repair or replace the fiber to resolve the issue.

Step 3: Step: Update the software or firmware, Command: upgrade system , Description: Update the software or firmware to resolve the issue.

Post-Check Verification

```
Command: display transceiver, Expected Output: Transceiver information, including temperature, voltage, and optical power levels., Description: Verify that the transceiver information is within normal operating ranges.

Command: display optical-link, Expected Output: Optical link information, including receive and transmit power levels, and fiber status., Description: Verify that the optical link information is within normal operating ranges.

Command: display system, Expected Output: System information, including software version and redundant link status., Description: Verify that the system information is within normal operating ranges.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore transceiver , Description: Restore the original SFP/XFP transceiver to resolve the issue.

- Step: Restore the original fiber, Command: restore fiber , Description: Restore the original fiber to resolve the issue.

- Step: Restore the original software or firmware, Command: restore system , Description: Restore the original software or firmware to resolve the issue.